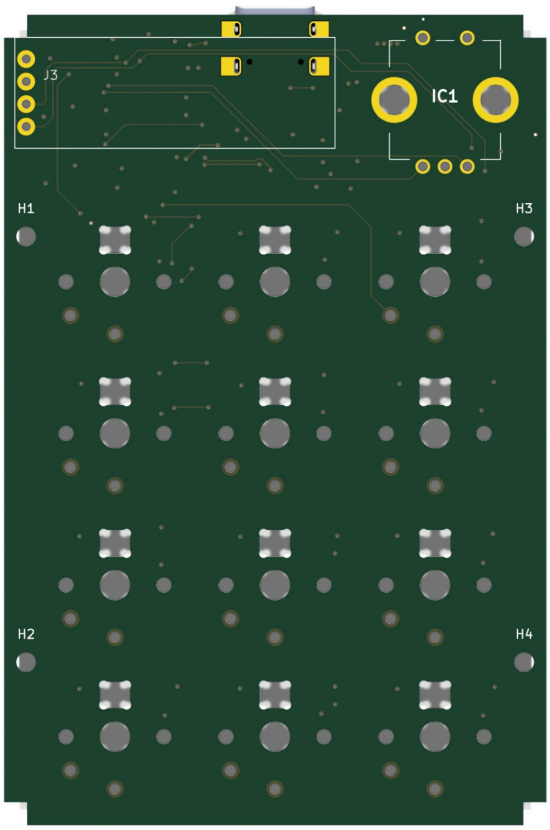
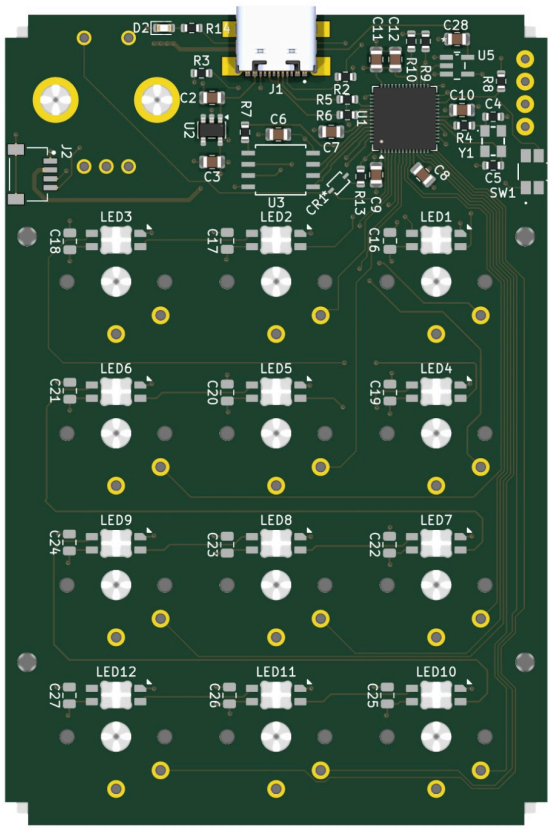
2040pad

BOARD	2040pad.kicad_pcb
REV	-
SIZE	61.1 x 91.5 mm
PARTS	32 placed, 39 DNP
DATE	2026-11-04



TOP SIDE



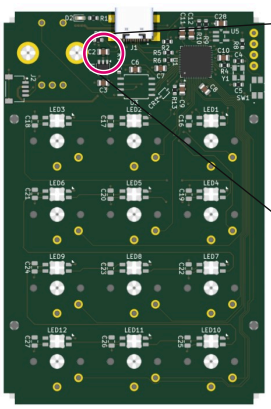
BOTTOM SIDE

Bill of materials

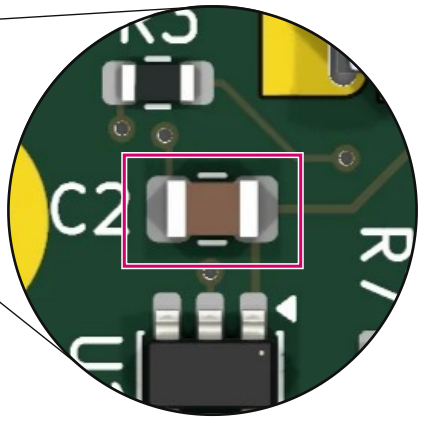
QTY	REFS	VALUE	PACKAGE	MPN	MAKER
2	C2, C3	10u	C_0805_2012Metric		
2	C4, C5	20p	R_0603_1608Metric		
8	C6, C7, C8, C9, C10, C11, C12, C28	1u	C_0805_2012Metric		
1	CR1	1N4148WS	SOD-323FL_ONS		
1	D2	LED	LED_0603_1608Metric		
1	J1	UJ20-C-H-G-SMT-1-P16-TR	SAMESKY_UJ20-C-H-G-SMT-1-P16-TR		Same Sky
6	R2, R3, R8, R9, R10, R14	5k1	R_0603_1608Metric		
2	R4, R13	1k	R_0603_1608Metric		
2	R5, R6	22	R_0603_1608Metric		
1	R7	10k	R_0603_1608Metric		
1	SW1	EVQ-P7J01P	SMD4_SW_EVQP7J_PAN		
1	U1	RP2040	QFN-56-1EP_7x7mm_P0.4mm_EP3.2x3.2mm		
1	U2	AP2112K-3.3	SOT-23-5		

QTY	REFS	VALUE	PACKAGE	MPN	MAKER
1	U3	W25Q64JVSSIM	SOIC_208MIL_WIN		
1	U5	SN74AHCT1G125DBVR	DBV5		
1	Y1	CX3225GB12000H0KPSC1	CX3225GB	CX3225GB12000H0KPSC1	Kyocera AVX
DO NOT PLACE	**C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27**	**1u**	**C_0805_2012Metric**		
DO NOT PLACE	**CH1, CH2, CH3, CH4, CH5, CH6, CH7, CH8, CH9, CH10, CH11, CH12**	**choc_v1_SW_solder**	**SW_choc_v1_1u**		
DO NOT PLACE	**IC1**	**PEC11R-4230K-S0024**	**PEC11R4230KS0024**	**PEC11R-4230K-S0024**	**Bourns**
DO NOT PLACE	**J2**	**SM04B-SRSS-TB_LF__SN_**	**JST_SM04B-SRSS-TB_LF__SN_**		**JST**
DO NOT PLACE	**J3**	**Conn_01x04_Socket**	**Waveshare .91 oled**		
DO NOT PLACE	**LED1, LED2, LED3, LED4, LED5, LED6, LED7, LED8, LED9, LED10, LED11, LED12**	**choc_SK6812MINI-E**	**LED_choc_6028R**		

Each step shows the whole board with a circled callout and an enlarged detail view of that circle. The part is outlined; where orientation matters, pin 1 is ringed and labelled. Detail scales are true when printed at 100%. The bottom side is shown as you see it when the board is flipped.



BOTTOM SIDE



DETAIL 1 · SCALE 6:1

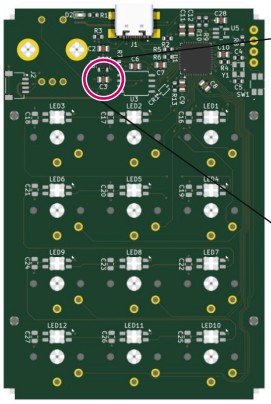
☐ **STEP 1 OF 32**

C2

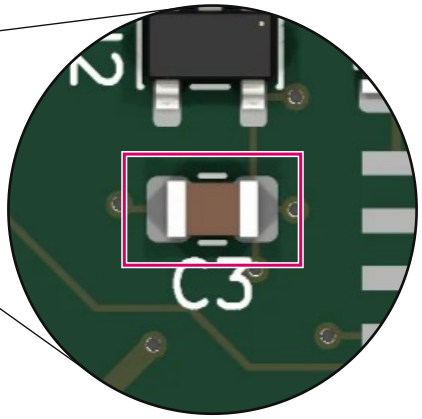
10u

Package C_0805_2012Metric

No Polarity



BOTTOM SIDE



DETAIL 2 · SCALE 6:1

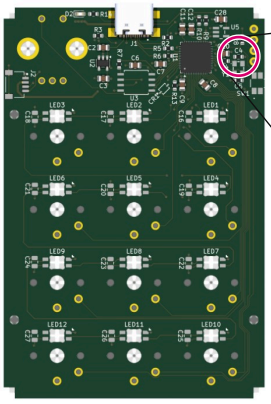
☐ **STEP 2 OF 32**

C3

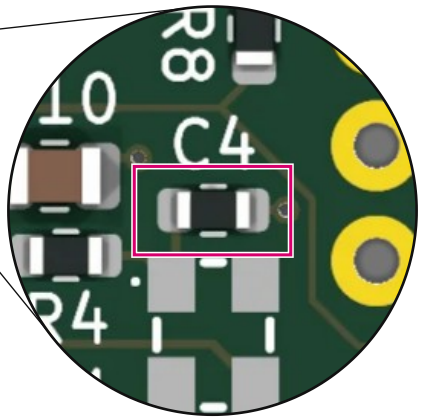
10u

Package C_0805_2012Metric

No Polarity



BOTTOM SIDE



DETAIL 3 · SCALE 6:1

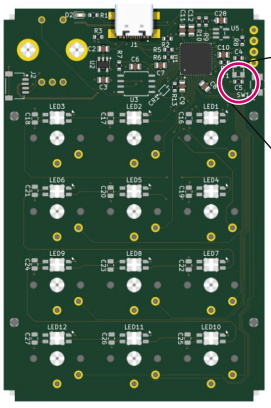
☐ **STEP 3 OF 32**

C4

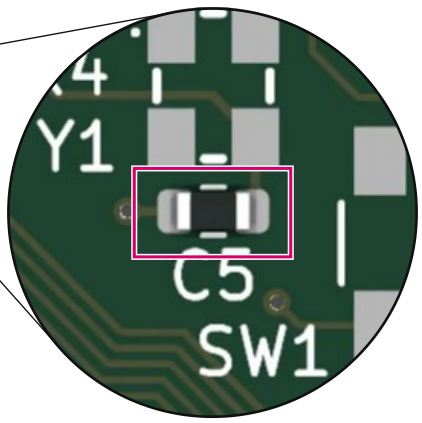
20p

Package R_0603_1608Metric

No Polarity



BOTTOM SIDE



DETAIL 4 · SCALE 6:1

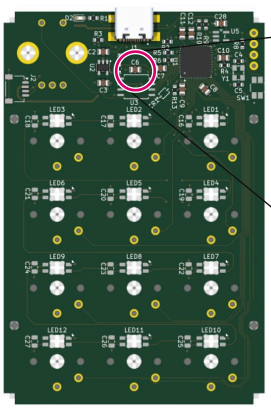
☐ **STEP 4 OF 32**

C5

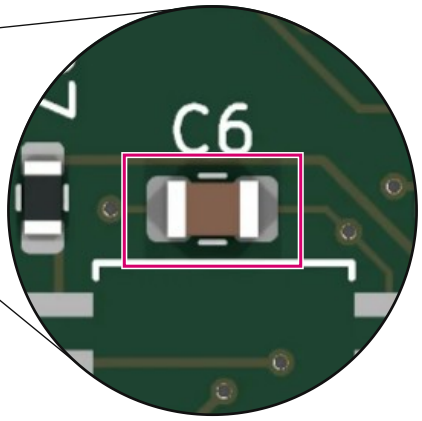
20p

Package R_0603_1608Metric

No Polarity



BOTTOM SIDE



DETAIL 5 · SCALE 6:1

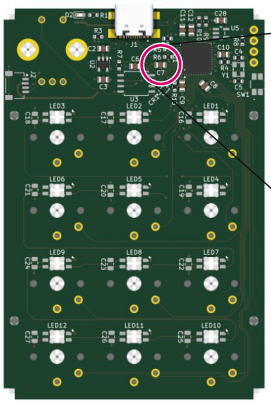
☐ **STEP 5 OF 32**

C6

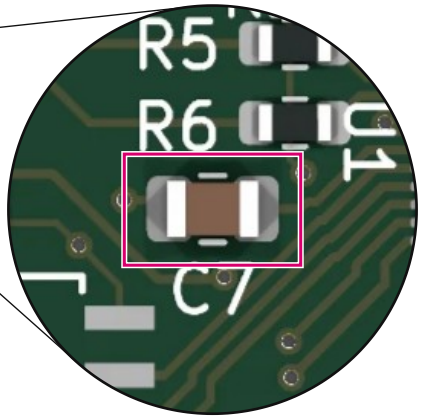
1u

Package C_0805_2012Metric

No Polarity



BOTTOM SIDE



DETAIL 6 · SCALE 6:1

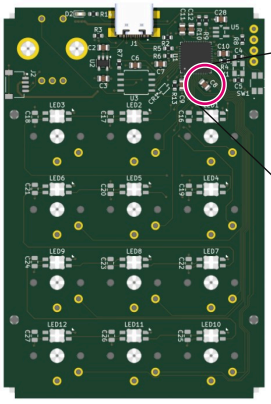
☐ **STEP 6 OF 32**

C7

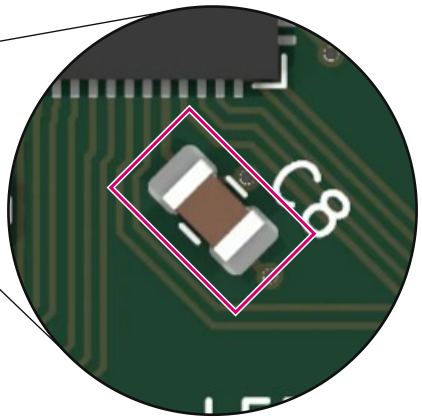
1u

Package C_0805_2012Metric

No Polarity



BOTTOM SIDE



DETAIL 7 · SCALE 6:1

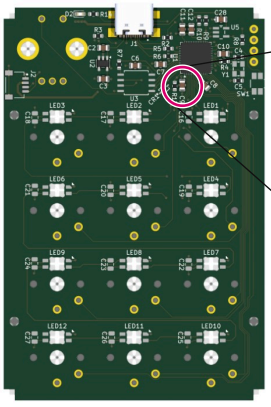
☐ **STEP 7 OF 32**

C8

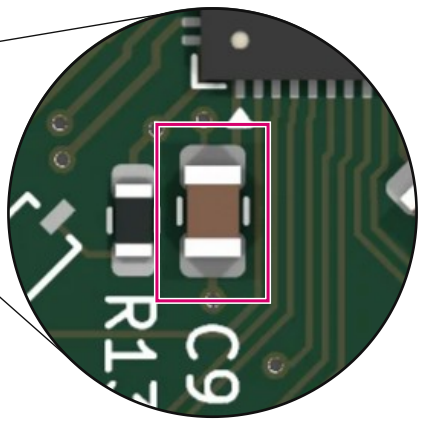
1u

Package C_0805_2012Metric

No Polarity



BOTTOM SIDE



DETAIL 8 · SCALE 6:1

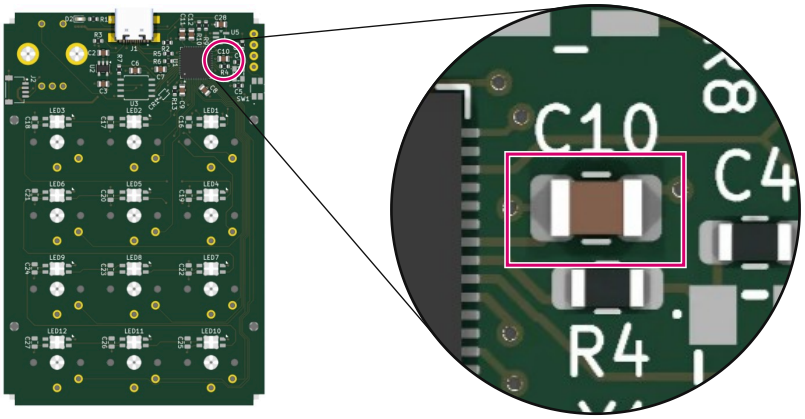
☐ **STEP 8 OF 32**

C9

1u

Package C_0805_2012Metric

No Polarity



BOTTOM SIDE

DETAIL 9 · SCALE 6:1

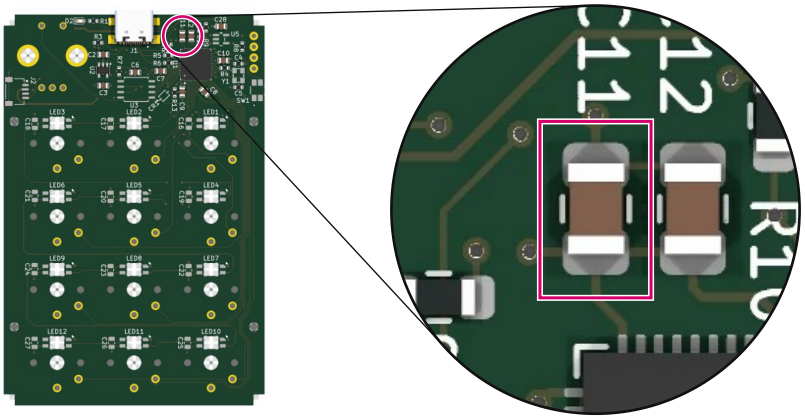
☐ **STEP 9 OF 32**

C10

1u

Package C_0805_2012Metric

No Polarity



BOTTOM SIDE

DETAIL 10 · SCALE 6:1

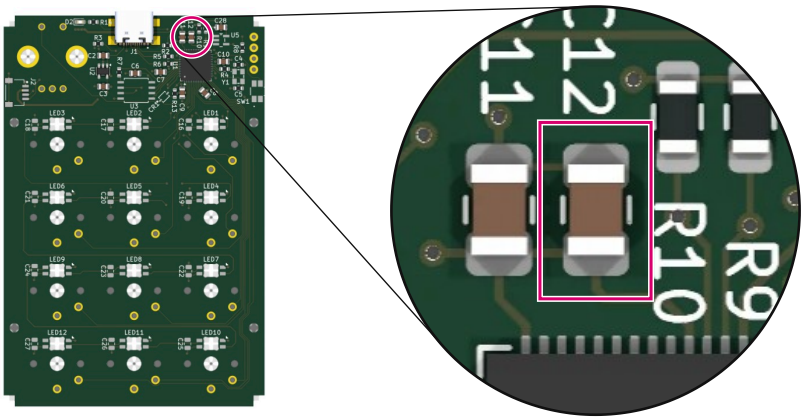
☐ **STEP 10 OF 32**

C11

1u

Package C_0805_2012Metric

No Polarity



BOTTOM SIDE

DETAIL 11 · SCALE 6:1

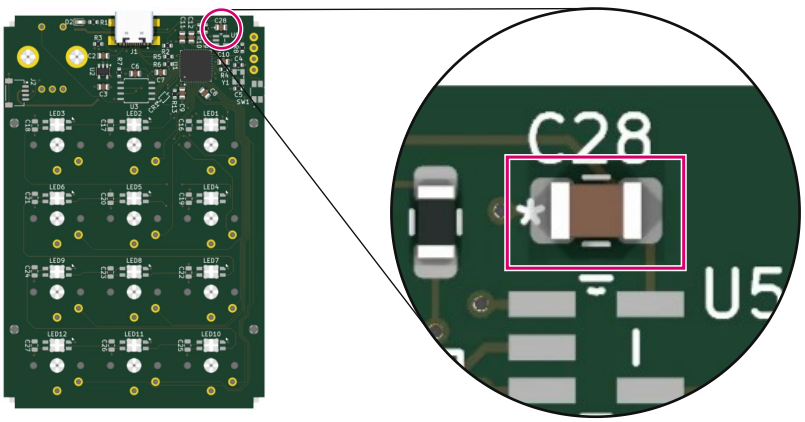
☐ **STEP 11 OF 32**

C12

1u

Package C_0805_2012Metric

No Polarity



BOTTOM SIDE

DETAIL 12 · SCALE 6:1

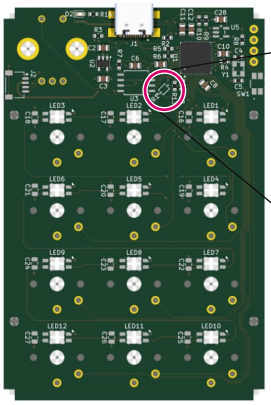
☐ **STEP 12 OF 32**

C28

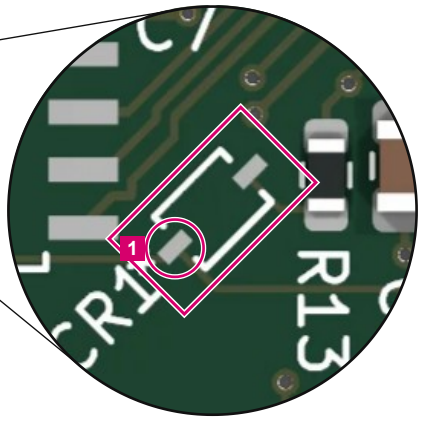
1u

Package C_0805_2012Metric

No Polarity



BOTTOM SIDE



DETAIL 13 · SCALE 6:1

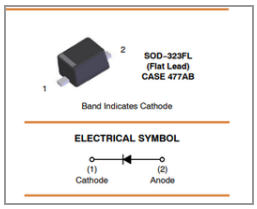
☐ **STEP 13 OF 32**

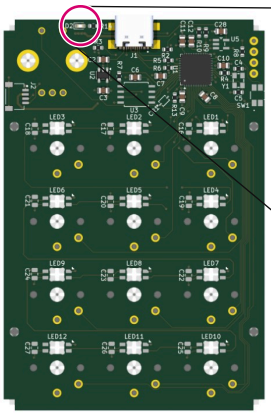
CR1

1N4148WS

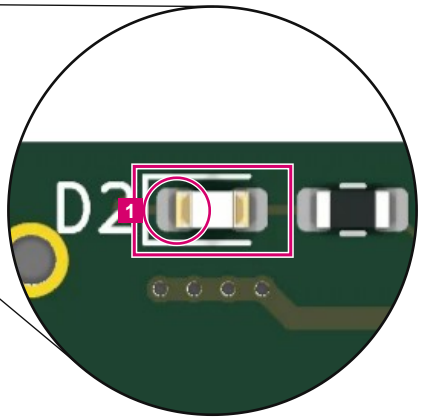
Package SOD-323FL_ONS

Device has polarity. The printed bar (pin 1) should face twoards the asterisk printed on the board near "CR1"





BOTTOM SIDE



DETAIL 14 · SCALE 6:1

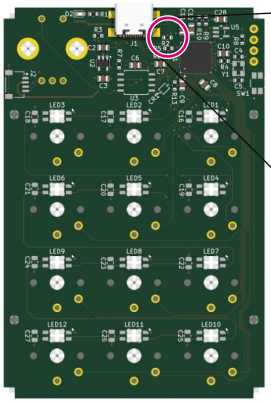
☐ **STEP 14 OF 32**

D2

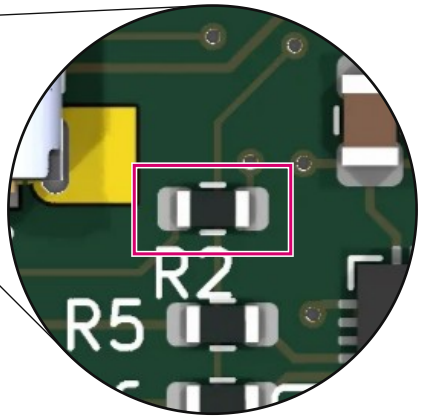
LED

Package LED_0603_1608Metric

Has Polarity! Cathode mark (green tick mark on bottom) should face the closed end of the 3 sided silk screen, or twoards D2



BOTTOM SIDE



DETAIL 15 · SCALE 6:1

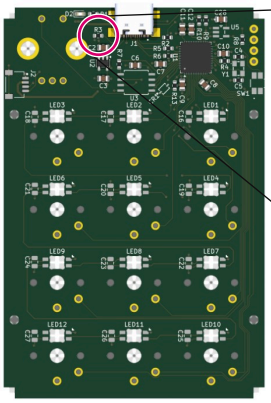
☐ **STEP 15 OF 32**

R2

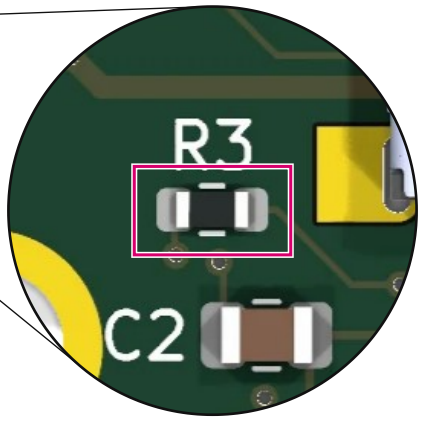
5k1

Package R_0603_1608Metric

No Polarity



BOTTOM SIDE



DETAIL 16 · SCALE 6:1

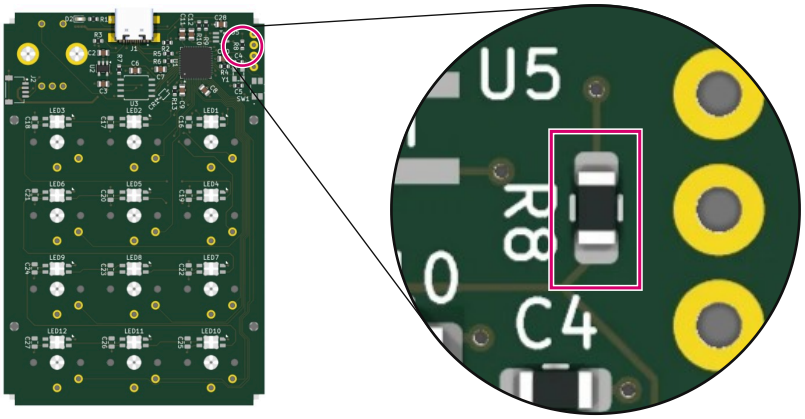
☐ **STEP 16 OF 32**

R3

5k1

Package R_0603_1608Metric

No Polarity



BOTTOM SIDE

DETAIL 17 · SCALE 6:1

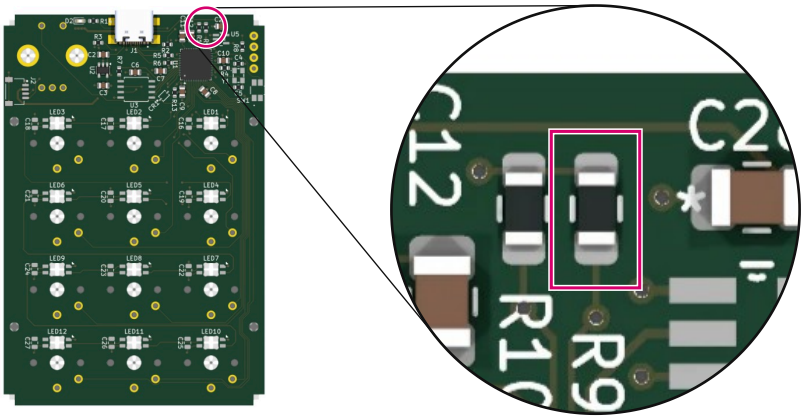
☐ **STEP 17 OF 32**

R8

5k1

Package R_0603_1608Metric

No Polarity



BOTTOM SIDE

DETAIL 18 · SCALE 6:1

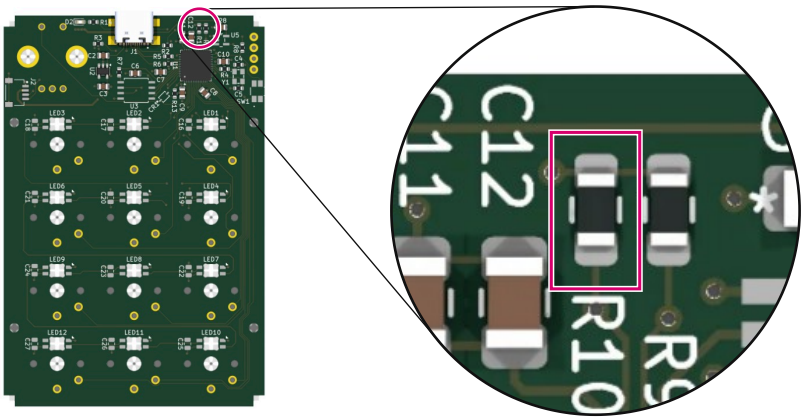
☐ **STEP 18 OF 32**

R9

5k1

Package R_0603_1608Metric

No Polarity



BOTTOM SIDE

DETAIL 19 · SCALE 6:1

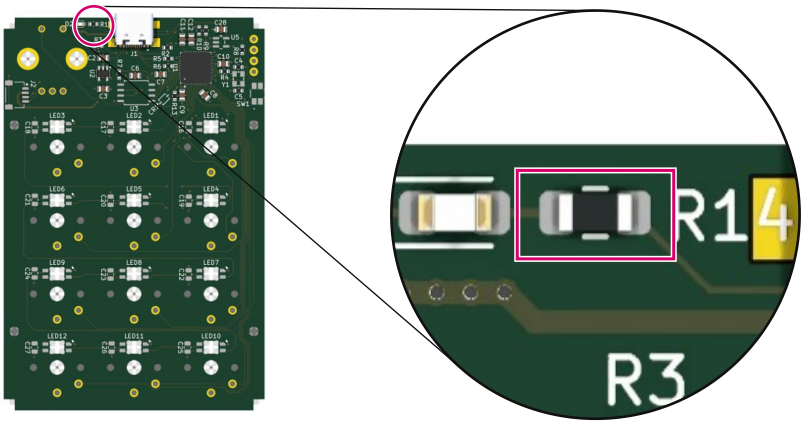
☐ **STEP 19 OF 32**

R10

5k1

Package R_0603_1608Metric

No Polarity



BOTTOM SIDE

DETAIL 20 · SCALE 6:1

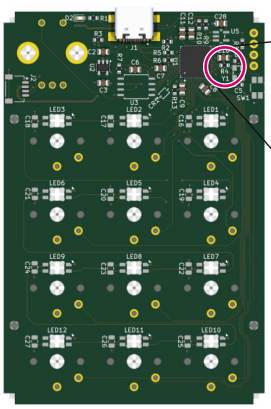
☐ **STEP 20 OF 32**

R14

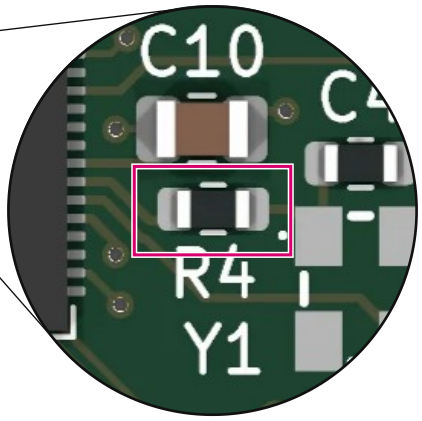
5k1

Package R_0603_1608Metric

No Polarity



BOTTOM SIDE



DETAIL 21 · SCALE 6:1

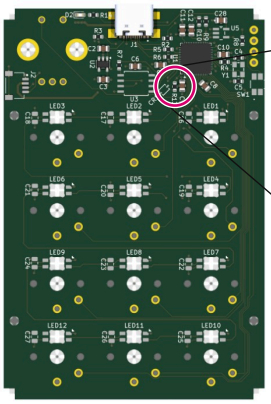
☐ **STEP 21 OF 32**

R4

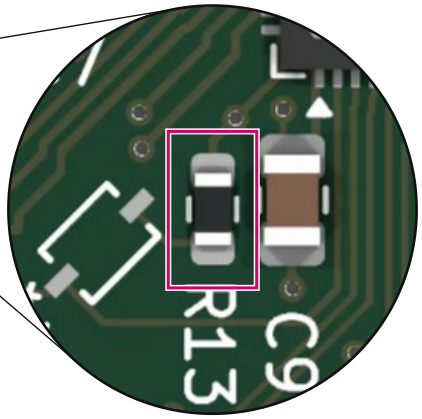
1k

Package R_0603_1608Metric

No Polarity



BOTTOM SIDE



DETAIL 22 · SCALE 6:1

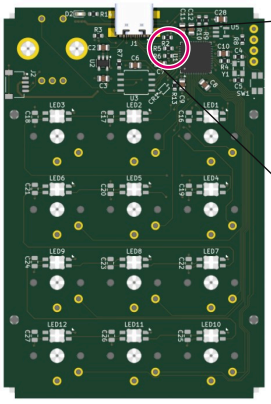
☐ **STEP 22 OF 32**

R13

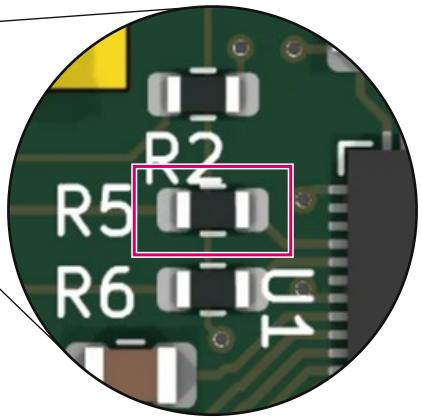
1k

Package R_0603_1608Metric

No Polarity



BOTTOM SIDE



DETAIL 23 · SCALE 6:1

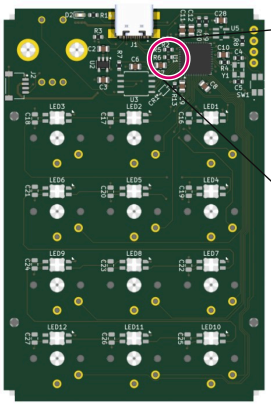
☐ **STEP 23 OF 32**

R5

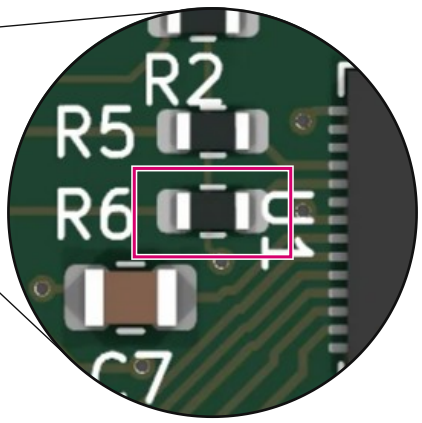
22

Package R_0603_1608Metric

No Polarity



BOTTOM SIDE



DETAIL 24 · SCALE 6:1

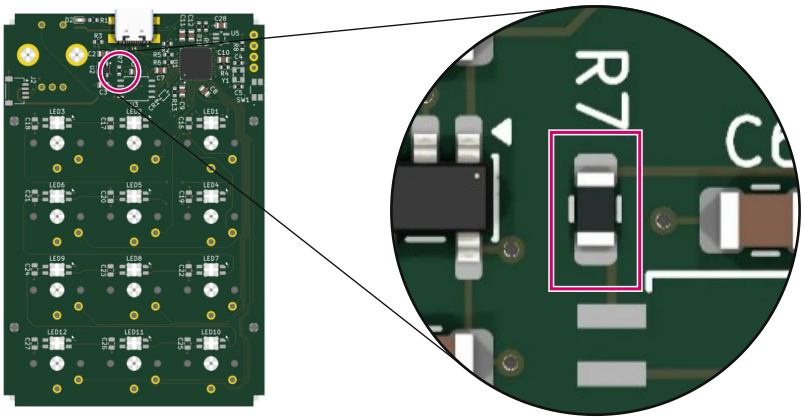
☐ **STEP 24 OF 32**

R6

22

Package R_0603_1608Metric

No Polarity



BOTTOM SIDE

DETAIL 25 · SCALE 6:1

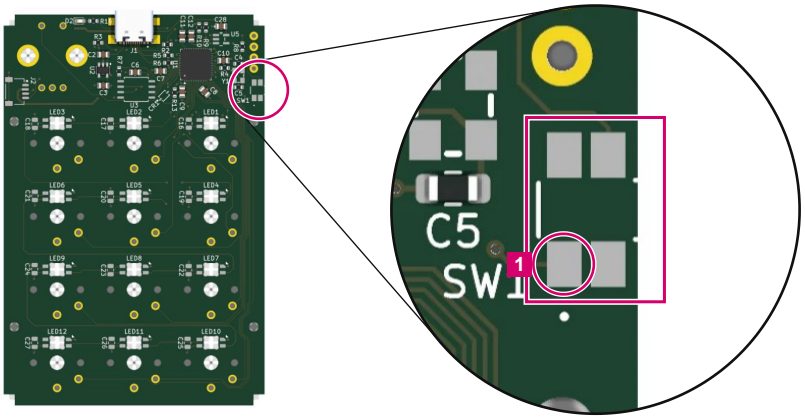
STEP 25 OF 32

R7

10k

Package R_0603_1608Metric

No Polarity



BOTTOM SIDE

DETAIL 26 · SCALE 4:1

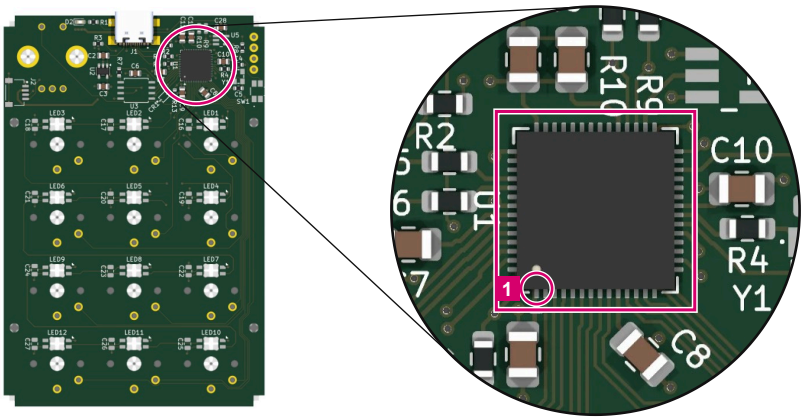
STEP 26 OF 32

SW1

EVQ-P7J01P

Package SMD4_SW_EVQP7J_PAN

place so button plunger faces out!



BOTTOM SIDE

DETAIL 27 · SCALE 3:1

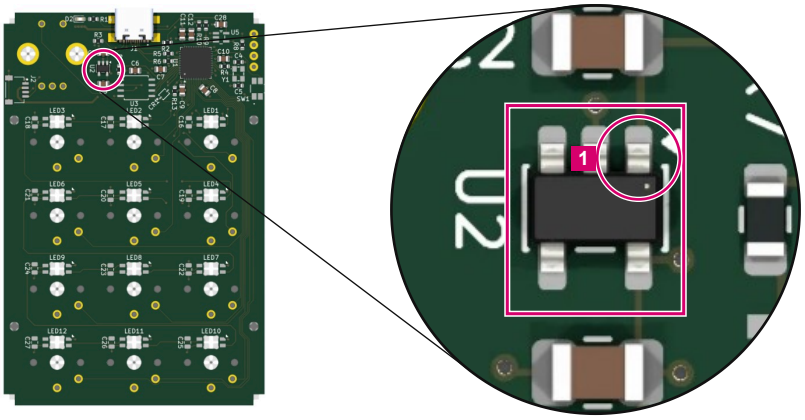
STEP 27 OF 32

U1

RP2040

Package QFN-56-1EP_7x7mm_P0.4mm_EP3.2x3.2mm

Has polarity! there should be a dot on one corner - this should face the arrow silk screened on the board, like in the image.



BOTTOM SIDE

DETAIL 28 · SCALE 6:1

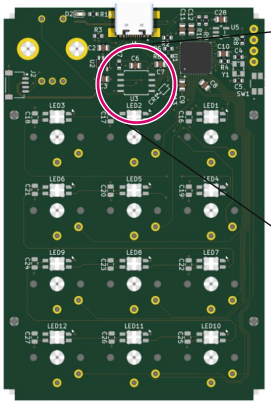
STEP 28 OF 32

U2

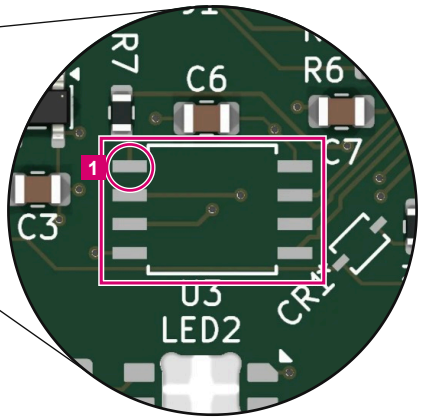
AP2112K-3.3

Package SOT-23-5

This one has a correct orientation but it is easy to tell - there are three legs on one side and two on the other. Match it up with the footprint.



BOTTOM SIDE



DETAIL 29 · SCALE 3:1

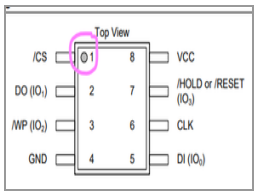
STEP 29 OF 32

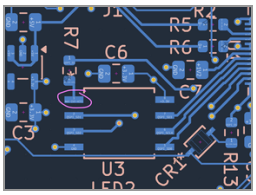
U3

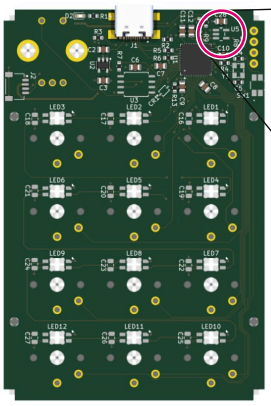
W25Q64JVSSIM

Package SOIC_208MIL_WIN

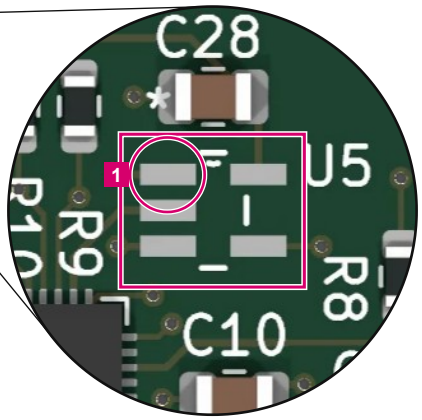
This part has a dot. The dot should be as far from the keys (face away from) the keys. The pin 1 silk is obscured by a part







BOTTOM SIDE



DETAIL 30 · SCALE 5:1

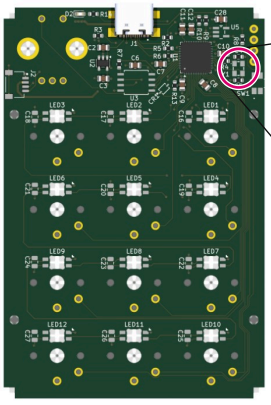
STEP 30 OF 32

U5

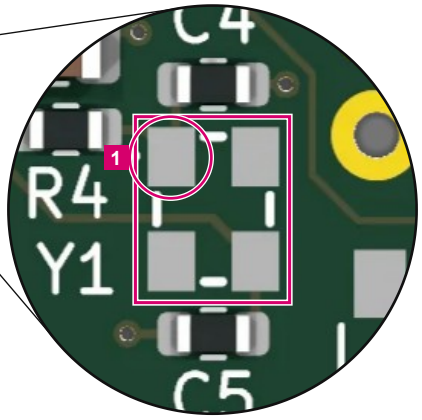
SN74AHCT1G125DBVR

Package DBV5

also polarized but hard to instlal backwards- line up 2 pin side with 2 pins on the board



BOTTOM SIDE



DETAIL 31 · SCALE 6:1

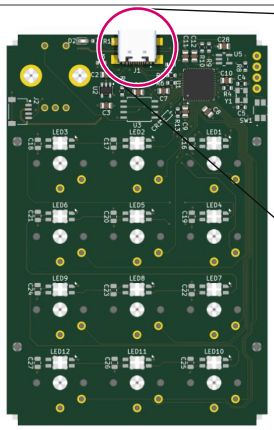
STEP 31 OF 32

Y1

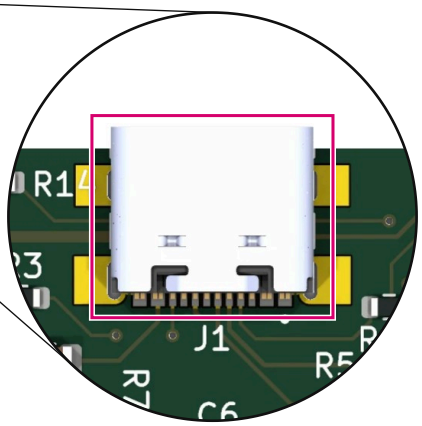
CX3225GB12000H0KPSC1

MPN CX3225GB12000H0KPSC1
Maker Kyocera AVX
Package CX3225GB

Crystal - not polarized, since it is rotationally symmetric.



BOTTOM SIDE



DETAIL 32 · SCALE 3:1

STEP 32 OF 32

J1

UJ20-C-H-G-SMT-1-P16-TR

Maker Same Sky
Package SAMESKY_UJ20-C-H-G-SMT-1-P16-TR

add thhis carefully and last!